The Cambridge capital controversy and two centers of gravitation: short-run partial-derivative analysis and long-run comparative statics

Minseong Kim

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Abstract

A change in the rate of interest moves every price in the economy, which critics in the Cambridge capital controversy took to rule out ceteris paribus and marginal-product scarcity reasoning about capital. We argue that the two sides (Cambridge, UK and Cambridge, US) in the controversy worked with different centers of gravitation. Within a short-run equilibrium, centered on the money rate of interest, a price-taking firm's partial-derivative conditions are valid and its demand for capital slopes downward. Comparisons of long-run positions with different uniform rates of interest are instead comparative statics, in which every price changes. The long-run neoclassical scarcity theory of capital needs regular technology to aggregate, and its normative use needs the long-run marginal product of capital to equal the rate of interest. That condition holds if and only if a change in the rate of interest leaves the value of the existing capital goods unchanged. For long-run distribution, a free market is therefore not enough even for those who accept distribution by marginal product as just.

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1 Introduction

Throughout the 1950s to the 1970s, economists at Cambridge, UK argued that the rate of interest acts on the economy through a more complicated channel than the neoclassical parable allows. In the parable, a lower rate of interest makes capital cheaper relative

to labor, firms adopt more capital-intensive techniques, and capital per worker rises. Robinson (1953) objected that capital cannot be measured independently of the rate of interest that capital is supposed to explain. Sraffa (1960) developed the objection by noting that the price of a capital good equals the wages paid for all labor used in producing the good, each accumulated with interest up to the date of sale. Labor applied earlier carries interest for longer, so a change in the rate of interest changes the prices of different capital goods by different amounts, and the relative prices of all capital goods shift. A lower rate of interest can then lead to a less capital-intensive technique (Garegnani, 1970, p. 423), and a technique abandoned at one rate can return at another, results that Samuelson (1966) accepted in his summing up of the debate. The underlying point was anticipated in the 1932 exchange between Sraffa and Hayek (Sraffa, 1932). An economy with many commodities has many commodity rates of interest, and these rates differ whenever relative prices change, so the rate of interest is not the uniform price of a single aggregate factor.

The neoclassical reply came through general equilibrium theory. In the sequential economy of Radner (1972), spot markets reopen at every date and agents trade a limited set of assets, with markets incomplete. Commodity rates of interest still differ across goods. Every intertemporal choice is nevertheless tied through the Euler equation to the money rate of interest, which in a modern economy is the rate that the central bank steers through its policy rate. Each commodity rate equals the money rate adjusted for the expected change in the money price of that commodity, and a central bank that targets inflation also fixes the expected path of a price index, so the real return measured in that index is a common center of gravitation for intertemporal choices. A firm deciding whether to hold more of a capital good compares the return on that good with this center of gravitation at the prices the firm faces. When each margin is examined separately and ceteris paribus, the marginal productivity theory of capital survives: at given prices, a lower cost of finance leads each firm to a more capital-intensive technique. The partial derivatives used in the ceteris paribus argument are defended as part of the definition of equilibrium. An equilibrium is a configuration from which no agent gains by deviating unilaterally, a condition that can alternatively be justified in a large economy where a single agent's deviation does not move prices. A plan that would pay only if prices moved is therefore counterproductive for the agent.

This paper offers an alternative understanding of what the controversy produced. The UK side concluded that neoclassical theory cannot support long-run equilibrium analysis of capital and distribution. The US side, and later general equilibrium theorists, concluded that general equilibrium theory, which does not need an aggregate of capital, had settled the controversy (Hahn, 1982a). We argue that this difference in conclusions has to do with different notions of the center of gravitation: the long-run uniform rate of interest for the UK side and the short-run money rate of interest for the US side. Much

of the UK critique was framed in terms of centers of gravitation (Garegnani, 1976; Petri, 2004), but the term can refer to positions of different duration. A long-run center of gravitation is the classical position with a uniform rate of profit, toward which market prices tend over long periods. A short-run center of gravitation is the rate around which intertemporal choices are organized at a given date. The neoclassical theory supplies a short-run center of gravitation, and in practice that center is the nominal interest rate that the central bank sets through the interest paid on reserves or through operations in government bonds. If the controversy concerned only whether a center of gravitation exists, the neoclassical concept of equilibrium could not be called problematic merely because an equilibrium requires a center of gravitation to be reachable. A controversy that remains unresolved must therefore concern another question: whether the short-run center of gravitation is enough for capital theory. The question is then whether the short-run neoclassical "scarcity" theory of capital, which involves first-order equilibrium conditions, needs to aggregate into the long-run neoclassical "scarcity" theory of capital, which involves comparative statics that compare long-run equilibrium positions at different rates of interest. The answer differs between the uses of the long-run theory in positive economics and in normative economics. As an approximation in positive economics, the long-run theory requires that technology be regular, and input-output evidence suggests that regularity usually holds. In normative economics, conclusions such as the claim that interest rewards the contribution of capital require the further condition $w'(r) = -k$, which is too restrictive to have any realistic possibility of being satisfied (Section 5).

Under the assumption of differentiability, neoclassical capital theory divides into two marginal product theories. The short-run theory holds at given prices, and the long-run theory compares long-run positions at different rates of interest. Making this distinction avoids much of the cross-talk that has kept the two sides from a meaningful dialogue. The defense through first-order conditions in a Radner economy is a defense of the short-run theory, and reswitching is a result about the long-run theory, so the two arguments do not contradict each other.

Equilibrium analysis usually assumes price-taking, and under that assumption the first-order conditions derived from partial derivatives give each firm a short-run downward-sloping demand for capital. In contrast, when two long-run positions are compared, every price differs, and the response of capital intensity contains a price-feedback term that the firm-level calculation leaves out. Section 4 explores this term with a well-known example from Samuelson (1966). The price-feedback term means that the long run need not follow the parable, in which a lower rate of interest reflects more abundant capital and goes with higher capital intensity. Section 5 shows that the long-run neoclassical "scarcity" theory needs two conditions. Regularity in the sense of Burmeister (1980) makes capital per worker fall as the rate of interest rises, and the available evidence suggests that regularity is the common case (Han and Schefold, 2006). The normative use of the theory depends

on identifying the rate of interest with the long-run marginal product of capital, and that identification needs the further condition $w'(r) = -k$, which holds if and only if a change in the rate of interest leaves the value of the existing capital goods unchanged (Section 5.1).

Aggregation in this paper is part of the long-run analysis. At a long-run position, the long-run rate of interest is the long-run center of gravitation and makes heterogeneous capital goods homogeneous over time: every capital good must earn that rate on its value, so capital can be treated as a single value magnitude. The comparative statics of the long-run theory then compare this magnitude across positions with different long-run rates of interest.

The short-run and long-run theories answer questions at different levels. The short-run theory concerns individual choices, the decisions of a firm or a household facing given prices. The short-run theory may also be used to justify a particular long-run equilibrium as the outcome of those choices. The long-run theory concerns how a society views its economy as a whole, for example whether an economy with a lower rate of interest has more capital per worker and how income is divided between wages and profits. Whether the interest and profits earned by capitalists are just therefore depends on the validity of the long-run theory. Clark (1899) presented marginal productivity as a natural law that would give every agent of production the wealth that agent creates if competition worked without friction, and the long-run parable makes the same claim by equating the rate of interest with the long-run marginal product of capital. That long-run marginal product is a comparative-statics result rather than a partial derivative. If the long-run marginal product of capital is no longer assured to equal the rate of interest, the theory can no longer support the claim that long-run distribution is just, socially speaking. The normative claim that distribution in a neoclassical free market is just has no general support in the theory.

When confronted with the critique of the normative marginal product claim, neoclassical economists may counter that the modern normative justification of neoclassical theory does not involve the marginal product of capital, so we briefly review that literature. It is clear from the start, however, that these defenses concern the short-run theory by design. The welfare theorems, core equivalence (Debreu and Scarf, 1963; Aumann, 1964) and the results reviewed below all evaluate a single equilibrium at given endowments, so they take the ownership of capital goods as given and do not compare long-run positions at different rates of interest. Makowski and Ostroy (2001, pp. 500–502) show that in large economies Walrasian allocations are the only ones in which each trader receives the marginal value of the trader's contribution to social surplus (Mas-Colell et al., 1995, Section 18.E), and Athreya (2013, p. 88) concludes that, when households start with similar endowments, Walrasian outcomes have a serious claim to being the only fair ones. The welfare theorems concern efficiency and, through lump-sum transfers, the

redistribution of endowments, but they say nothing about whether the long-run distribution of income between wages and interest is just, and in economies with overlapping generations and capital a competitive equilibrium can accumulate too much capital and fail to be efficient (Diamond, 1965). None of these results shows that the rate of interest equals the long-run marginal product of capital, which is the traditional basis for the justice of interest.

We would like to note that this paper does not claim to represent the whole of the Cambridge capital debate. The paper aims to clarify the most controversial and likely the simplest aspect of the debate: whether the marginal productivity theory of capital can rely on partial derivatives and ceteris paribus reasoning. Other parts of the debate, such as the alternative theories of distribution developed on the Cambridge, UK side, are outside the scope of this paper.

Section 2 sets out the sequential reply and its short-run center of gravitation. Section 3 presents the short-run theory, Section 4 the long-run comparison, and Section 5 aggregation and the long-run marginal product of capital.

2 The sequential neoclassical reply and the short-run center of gravitation

After 1966 the neoclassical defense moved to general equilibrium, where capital is a vector of physically distinct goods with separate prices and rentals (Hicks, 1939; Debreu, 1959; Arrow and Debreu, 1954). In the sequential version of Radner (1972), let $q_{l,t}$ be the spot price of good l at date t in units of account, and let a safe monetary asset bought for one unit of account at date 0 pay R units at date 1. The return R is the gross money rate of interest, the rate a central bank steers through its policy rate. At an interior optimum a consumer's marginal rate of substitution between good l at the two dates satisfies the Euler equation

$$\frac{\partial U/\partial x_{l,0}}{\partial U/\partial x_{l,1}} = R\,\frac{q_{l,0}}{q_{l,1}}. \tag{1}$$

The right-hand side is the commodity rate of interest of good l, equal to the money rate adjusted for the expected change in the money price of good l. Commodity rates therefore differ across goods whenever relative prices change, as Sraffa said. Every commodity rate is nevertheless tied to R. A capital good that costs $q_{k,0}$, earns a rental f_k and depreciates at rate δ_k is held only if

$$R = \frac{f_k + (1-\delta_k)\,q_{k,1}}{q_{k,0}}, \tag{2}$$

which is the user cost condition of Jorgenson (1963). The money rate is thus the short-run center of gravitation against which every firm values investment, and under an inflation target the real rate in units of the targeted price index plays the same role. In practice the

central bank sets this rate through the interest paid on reserves and through operations in government bonds. With incomplete markets the center of gravitation pins down values only for payoffs that traded assets span (Magill and Shafer, 1991), but every firm values the spanned part of an investment against the same rate.

Garegnani (1976) replied that the move to general equilibrium changed the notion of equilibrium. An equilibrium built on an arbitrary initial vector of capital goods lacks the persistence of the classical long-period position, because investment alters the vector and the equilibrium changes with the vector. Garegnani concluded that the neo-Walrasian equilibrium could not be a center of gravitation for actual prices. The argument concerns the long-run center of gravitation. The neo-Walrasian equilibrium still has a short-run center of gravitation in the money rate described above, so the persistence argument alone does not show that the neoclassical concept of equilibrium is problematic.

3 The short-run theory: partial-derivative unilateral-deviation analysis

Within a short-run equilibrium the first-order conditions of firms are partial derivatives at given prices, as ordinary price-taking implies. A single firm in a large competitive economy does not move the rate of interest or the prices of capital goods, so any change of plan by that firm is evaluated at equilibrium prices, and a plan that would pay only if prices moved is counterproductive to the firm.

The term short-run refers here to what is held given and says nothing about the length of the horizon. An Arrow-Debreu or Radner equilibrium can extend over many dates and still describe equilibrium at a given initial endowment, so the short-run theory covers intertemporal equilibrium and the dynamics of a sequential economy along a given path. The long-run theory of Section 4 focuses on the long-run tendency of an economy or a society, whereas the short-run theory explains how each long-run path may be reached from the collection of individual choices.

Let each technique τ in a set $\mathcal{T}$ require l_τ units of labor and the vector $\mathbf{k}_\tau$ of capital goods per unit of output, with input vector $z_\tau = (l_\tau, \mathbf{k}_\tau)$. Let input prices be $\pi = (w, (1+r)\mathbf{p})$, where w is the wage and $(1+r)\mathbf{p}$ is the cost of the services of capital goods advanced for one period.

The next proposition concerns a single firm that faces a given wage and given prices of capital goods, and asks how the firm's choice of technique responds to the rate of interest the firm pays.

Proposition 1. *Hold the wage w and the prices $\mathbf{p}$ of capital goods fixed.*

(i) The firm's demand for capital slopes downward: if the rate of interest the firm pays

falls from r to r', the value at prices $\mathbf{p}$ of the capital goods used per unit of output does not fall (Varian, 1984).

(ii) No reswitching occurs in the firm's choice: for each technique τ, the set of input prices at which τ minimizes unit cost, $C_\tau = \{\pi : z_\tau \cdot \pi \leq z_\sigma \cdot \pi \text{ for all } \sigma \in \mathcal{T}\}$, is a convex cone (Mas-Colell et al., 1995, Proposition 5.C.2(i)–(ii)), so each technique minimizes cost on an interval of rates of interest, and a technique abandoned as the rate falls cannot return.

Proof. (i) Let τ be chosen at r and τ' at $r' < r$, with $\pi(r) = (w, (1+r)\mathbf{p})$. Cost minimization at both price vectors gives $(z_{\tau'} - z_\tau) \cdot (\pi(r') - \pi(r)) \leq 0$, which reduces to $(r' - r)\,\mathbf{p} \cdot (\mathbf{k}_{\tau'} - \mathbf{k}_\tau) \leq 0$ and hence $\mathbf{p} \cdot \mathbf{k}_{\tau'} \geq \mathbf{p} \cdot \mathbf{k}_\tau$. (ii) Let $c(\pi) = \min_{\sigma \in \mathcal{T}} z_\sigma \cdot \pi$ be the unit cost function, so that $C_\tau = \{\pi : z_\tau \cdot \pi - c(\pi) \leq 0\}$. The cost function is homogeneous of degree one and concave in input prices (Mas-Colell et al., 1995, Proposition 5.C.2(i)–(ii)), so $z_\tau \cdot \pi - c(\pi)$ is homogeneous of degree one and convex, and the set on which this function is not positive is a convex cone. As r varies, $\pi(r) = (w, \mathbf{p}) + r\,(0, \mathbf{p})$ moves along a straight line, and the intersection of a line with a convex set is an interval. This is the short-run ceteris paribus result. □

Price-taking establishes that an equilibrium is immune to unilateral deviations, but price-taking says nothing about whether the equilibrium is unique or whether a process of price adjustment would reach that equilibrium. How an economy reaches equilibrium remains an unsettled question in general equilibrium theory (Hahn, 1982b; Fisher, 1983). The simplest of the examples in Scarf (1960, pp. 161–163) is an exchange economy with three consumers and three goods, in which each consumer owns one good and wants two goods in fixed proportions. In this economy, now known as Scarf's economy, the tâtonnement process, where each price rises with the excess demand for its good, cycles without converging. The instability does not depend on the extreme features of this example, because Scarf (1960, Section 3) obtains the same result with smooth, strictly convex preferences. The results of Sonnenschein (1973) and Debreu (1974) show that individual rationality places almost no restriction on market excess demand, so stability cannot be derived from optimizing behavior alone. Saari and Simon (1978) showed that price-adjustment processes that always converge must use information on how each excess demand responds to every price. Fisher (1983) studied processes in which agents trade out of equilibrium, and he could prove stability only under the further assumption that agents meet no favorable surprises. Experimental and game-theoretic work has since qualified these results. In laboratory double auctions, prices in Scarf's economy follow the paths that the tâtonnement model predicts, converging where the model predicts convergence and orbiting where the model predicts orbits (Anderson et al., 2004), and a market institution built on Smale's global Newton method makes prices in Scarf's economy converge

to the competitive equilibrium (Goeree and Lindsay, 2016). Strategic market games and bargaining models show that equilibria of decentralized trade implement Walrasian allocations (Shapley and Shubik, 1977; Gale, 1986; Giraud, 2003). Market games, however, assume a centralized rule that sets prices from aggregate bids, and bargaining models deliver Walrasian outcomes only when traders remain anonymous (Athreya, 2013, p. 87). Athreya (2013, pp. 93–96), writing in defense of Walrasian macroeconomics, concludes that experiments show what markets do more clearly than why they do it. Apart from Fisher (1983), whose full model adds firms and production, the results reviewed above concern pure exchange economies without produced inputs. Garegnani (2000) argued that intertemporal equilibria depend on a value notion of capital through the market for savings, Schefold (2005) that reswitching can make them unstable, and Mandler (2005) that they are well behaved under conditions such as a single consumer, a claim Garegnani (2005) disputed. While the short-run theory is secure in characterizing an equilibrium, its account of how the economy reaches an equilibrium remains unclear.

4 The long-run theory: comparative statics analysis

The long-run analysis sets an economy in a long-run position at one rate of interest beside an economy at another rate and asks how their techniques and capital intensities differ. In this comparison, no firm can deviate unilaterally, and every firm deviates to another long-run equilibrium position.

We first take a fixed technology, given by a single input matrix A, so that a change in the rate of interest alters relative prices without altering the methods of production. We expect the comparison to depart further from the parable when the technology is flexible, because the change in relative prices then also changes which technique is cheapest. The example of Samuelson (1966) in Section 4.2 confirms this expectation with reswitching.

4.1 Fixed input-output matrix

To see how much prices change with the rate of interest, take an economy that produces n goods with circulating capital, meaning that produced inputs are used up within one period of production. The input matrix $A \geq 0$ has entry a_{jk} equal to the amount of good k used to produce one unit of good j, so the jth row a_j lists the produced inputs of industry j. We assume that A is productive, meaning that the economy can produce more of every good than it uses up as inputs, and indecomposable, meaning that every good enters directly or indirectly into the production of every other good. In the terms of Sraffa (1960), every good is then a basic commodity. The direct labor requirement $l_j > 0$ is the labor used in industry j to produce one unit of good j, without counting the labor used to produce the inputs of industry j. These terms are standard in input-

output analysis. We take the wage as numeraire, so every price is measured in wage units. Competitive prices then satisfy $\mathbf{p} = (1+r)A\mathbf{p} + l$: each price covers the cost of produced inputs with one period of interest, plus the direct wage bill. Therefore

$$\mathbf{p}(r) = \sum_{t=0}^{\infty} (1+r)^t A^t l \tag{3}$$

for $1+r$ below the reciprocal of the dominant eigenvalue of A. Equation (3) reduces each price to dated quantities of labor. The vector l is the labor applied in the final period of production, Al is the labor that produced the inputs used in the final period, $A^2 l$ is the labor that produced those inputs, and so on, so $A^t l$ is the labor applied t periods before the output. The wages for this labor were paid t periods before the sale and carry interest until the sale, so this labor adds $(1+r)^t [A^t l]_j$ to the price of good j. Let

$$s_{jt}(r) = \frac{(1+r)^t [A^t l]_j}{p_j(r)}$$

be the share of the price of good j that pays for labor applied t periods earlier, including interest. These shares are nonnegative and sum to one over t, and $s_{j0} = l_j / p_j(r)$ is the share of direct labor. The average period of production of good j is

$$T_j(r) = \sum_t t\, s_{jt}(r),$$

the average time between applying labor and selling the output of good j, with each date weighted by its share in the price. Equation (3) holds only at a long-run equilibrium and does not describe prices in general, because the equation requires every industry to earn the same rate of interest on the value of its produced inputs. Proposition 2 and the elasticity results that follow are therefore comparative statics results that compare long-run positions at different rates of interest.

The next proposition explains why partial derivatives and ceteris paribus reasoning cannot be applied in the long-run theory: by differentiating Equation (3) with respect to the rate of interest, we can show that a change in the rate of interest always changes competitive prices in wage units, and even if we insist on taking prices as fixed, those prices produce nonzero economic profits or losses for firms, whereas a competitive equilibrium requires zero economic profit or loss.

Proposition 2 (Hicks, 1939, pp. 186–187; Sraffa, 1960, Chapter VI)**.** *The elasticity of p_j with respect to $1+r$ equals $T_j(r)$, and $T_j(r) \geq 1 - l_j/p_j(r)$, where $1 - l_j/p_j(r)$ is the share of produced inputs in the cost of good j. If the rate of interest changes from r to r' while prices stay at $\mathbf{p}(r)$, every industry j that uses produced inputs makes a profit or loss of $(r - r')\, a_j \cdot \mathbf{p}(r)$ per unit, where a_j is the jth row of A.*

Proof. Differentiating (3) term by term gives $(1+r)\,\partial p_j/\partial(1+r) = \sum_t t(1+r)^t[A^t l]_j$, and dividing by p_j gives T_j. Every term with $t \geq 1$ has weight at least one, so $T_j \geq 1 - s_{j0} = 1 - l_j/p_j$. At prices $\mathbf{p}(r)$ and rate r', industry j earns $p_j - (1+r')\,a_j \cdot \mathbf{p} - l_j$ per unit, and substituting $p_j = (1+r)\,a_j \cdot \mathbf{p} + l_j$ gives $(r - r')\,a_j \cdot \mathbf{p}$. □

4.2 Flexible input-output matrix

A possible objection to Section 4.1 is that prices move with the rate of interest only because the input matrix is fixed. With a flexible technology, firms might change technique as the rate of interest changes, so that prices could stay at $\mathbf{p}(r)$ at another rate r'. Two short arguments show that flexibility does not rescue fixed prices.

Let each industry j choose from a set of techniques, finite or infinite, and let technique τ use l_τ units of labor and the vector a_τ of produced inputs per unit of output. We assume that every technique uses some produced input, which extends the indecomposability of Section 4.1 to a flexible technology. With the wage as numeraire, the unit cost of technique τ at the rate of interest r and prices $\mathbf{p}$ is $l_\tau + (1+r)\,a_\tau \cdot \mathbf{p}$. Hold prices at $\mathbf{p}(r)$ and raise the rate of interest to $r' > r$. For every technique τ of industry j,

$$l_\tau + (1+r')\,a_\tau \cdot \mathbf{p}(r) \;>\; l_\tau + (1+r)\,a_\tau \cdot \mathbf{p}(r) \;\geq\; p_j(r),$$

where the first inequality holds because the interest charge rises and nothing else changes, and the second holds because $p_j(r)$ is the lowest unit cost at r. Industry j therefore makes a loss at r' whichever technique the industry chooses. For $r' < r$, the technique in use at r becomes strictly cheaper at r', so the industry makes a profit. Prices fixed at $\mathbf{p}(r)$ cannot belong to a long-run position at any other rate of interest, whatever the set of techniques, and the second part of Proposition 2 extends from a fixed input matrix to any technology.

A weaker form of the objection holds that flexibility at least reduces the response of prices to the rate of interest. To first order, flexibility does not reduce that response. Assuming that techniques form a continuum, let $c_j(\pi)$ be the unit cost function of good j at input prices $\pi(r) = (1, (1+r)\mathbf{p}(r))$, so that long-run prices satisfy

$$p_j(r) = c_j\big(1, (1+r)\mathbf{p}(r)\big), \qquad j = 1, \ldots, n. \tag{4}$$

We assume that each c_j is twice differentiable. By Shephard's lemma (Shephard, 1953; Mas-Colell et al., 1995, Proposition 5.C.2(vi)), $\partial c_j/\partial \pi_s = a_{js}$ for each produced input s and $\partial c_j/\partial w = l_j$ for labor. The derivative $\mathbf{p}'(r)$ can be obtained in two ways.

The first way differentiates Equation (4) directly with respect to r. The wage is fixed

at one, and the rental of produced input s is $\pi_s = (1+r)\,p_s(r)$, so the chain rule gives

$$p_j'(r) = \sum_s \frac{\partial c_j}{\partial \pi_s}\,\frac{d\pi_s}{dr} = \sum_s \frac{\partial c_j}{\partial \pi_s}\,\big[p_s(r) + (1+r)\,p_s'(r)\big].$$

Shephard's lemma then replaces each $\partial c_j/\partial \pi_s$ with a_{js}, which gives $p_j'(r) = \sum_s a_{js}\,[p_s(r) + (1+r)\,p_s'(r)]$, or $\mathbf{p}'(r) = A\,[\mathbf{p} + (1+r)\mathbf{p}'(r)]$ in matrix form, where A is the input matrix of the techniques in use at r.

The second way first rewrites Equation (4). Because c_j is homogeneous of degree one, Euler's theorem gives $c_j(\pi) = \sum_i \pi_i\,\partial c_j/\partial \pi_i$, and Shephard's lemma turns this sum into $l_j + (1+r)\sum_s a_{js}\,p_s$. Equation (4) therefore becomes

$$\mathbf{p}(r) = (1+r)\,A(r)\,\mathbf{p}(r) + l(r),$$

which is the price equation of Section 4.1 with the input matrix and labor requirements of the techniques in use at r. Differentiating this equation in matrix form gives

$$\mathbf{p}'(r) = A\,\mathbf{p} + (1+r)\,A\,\mathbf{p}'(r) + \big[(1+r)\,A'(r)\,\mathbf{p} + l'(r)\big].$$

The bracketed term is zero: each technique in use minimizes cost, so a small change of technique leaves unit cost unchanged to first order. Both ways give $\mathbf{p}'(r) = A\,[\mathbf{p} + (1+r)\mathbf{p}'(r)]$, and hence

$$\mathbf{p}'(r) = [I - (1+r)A]^{-1} A\,\mathbf{p}. \tag{5}$$

Equation (5) is the same as the derivative of Equation (3) for a fixed matrix equal to the techniques in use at r, and Proposition 2 holds at every rate of interest. Opocher and Steedman (2015, p. 95) make a related point: the relative prices of produced inputs cannot be treated as given parameters when long-run positions are compared.

The response of capital intensity can be described with the same tools. For one good, let $(l, \mathbf{k}) = \nabla c(\pi)$ be the input requirements per unit of output, where l is labor and $\mathbf{k}$ is the vector of capital goods, and let S be the Hessian of c, which is symmetric and negative semidefinite because c is concave (Mas-Colell et al., 1995, Proposition 5.C.2(vii)). Following the method of Steedman (2005) and Opocher and Steedman (2015, Section 4.8), who apply the Hessian of the cost function to the change in all input prices together, we write $d\pi/dr = u + (1+r)v$ with $u = (0, \mathbf{p})$ and $v = (0, \mathbf{p}'(r))$. The direction u is the rise in the interest charge on every capital good at given prices, and the direction v is the change in the prices of capital goods themselves. The input requirements change by $S\,d\pi/dr$, and valuing the change in capital goods at current prices means multiplying by $u^\top$, because u has the price p_s in the entry of capital good s and zero in the entry of labor. The change in the value of capital per unit of output, measured at current prices,

is therefore

$$\mathbf{p} \cdot \frac{d\mathbf{k}}{dr} = u^\top S u + (1+r)\, u^\top S v. \tag{6}$$

We call the term $u^\top S u$ the substitution term. The substitution term is the response of a firm that sees the interest charge change while the prices of capital goods stay fixed, which is the firm-level response of Proposition 1, and the term is never positive because S is negative semidefinite. We call the term $(1+r)\, u^\top S v$ the price-feedback term. The term records how the change in capital-good prices alters the choice of technique, and its sign depends on the technology.

With a finite set of techniques, the unit cost function has kinks, and Equation (6) does not apply. An example of Samuelson (1966) shows the outcome in that case. Technique A applies 7 units of labor two periods before output, and technique B applies 2 units three periods before and 6 units one period before. With the wage as numeraire, unit costs are $7(1+r)^2$ and $2(1+r)^3 + 6(1+r)$. They are equal at $r = 0.5$ and at $r = 1$, with A cheaper below 50 percent, B between 50 and 100 percent, and A again above. In a stationary state producing one unit per period, the output price p covers wages and profits, $p = L + rK$, so the value of capital per worker is $k = (p - L)/(rL)$. Table 1 evaluates it at the two switch points, where both techniques are compared at the same prices.

Switch point	Output price	k for A	k for B	Change as r rises
$r = 0.5$	15.75	2.50	1.94	A to B, k falls
$r = 1.0$	28.00	3.00	2.50	B to A, k rises

Table 1: Capital per worker, in wage units, at the switch points of the example of Samuelson (1966).

At the first switch the economy behaves as the parable says, and capital per worker falls as the rate of interest rises. At the second switch capital per worker rises with the rate of interest, despite each firm minimizing cost at the prices it faces and Proposition 1 holding for every firm at every rate of interest.

5 Aggregation and the long-run marginal product of capital

Neoclassical economists may object that aggregation is beside the point, because neoclassical theory, and general equilibrium theory in particular, does not require an aggregate of capital. The objection is correct for the short-run theory of Section 3, which works with vectors of heterogeneous capital goods and their prices, but aggregation here is part of the long-run analysis. At a long-run position, competition moves capital toward the

industries with the highest returns until every industry earns the same rate of interest on the value of its capital goods. With a uniform rate of interest, a unit of value invested in any capital good earns the same return, so the heterogeneous capital goods form a single aggregate of value. Aggregation therefore follows from the long-run center of gravitation.

5.1 Theory

The neoclassical aggregate theory of capital asserts a downward-sloping relation between the rate of interest and capital per worker across long-run positions, from which the aggregate production function, the marginal productivity theory of distribution and the identification of the rate of interest with the scarcity of capital all follow. Section 4 shows that the behavior of the firms that make up the economy does not by itself imply this relation. The aggregate demand for capital can slope up even when every firm's demand for capital slopes down.

The aggregate marginal productivity theory requires a condition that Burmeister (1980) calls regularity: for the economy as a whole, the price-feedback effects of Equation (6) must be small enough relative to the substitution effects that the real Wicksell effect is negative. Regularity is a condition on the available techniques, given the composition of net output, and does not follow from optimizing behavior. We use the term in Burmeister's technical sense, as a sign condition on the real Wicksell effect, and attach no normative meaning: Pasinetti (1979, p. 655) objects that the term suggests technologies with reswitching are somehow abnormal. Regularity means that, valued at the prices of a given long-run position, the capital goods per worker fall as the rate of interest rises. Net output per worker then falls as well, as Equation (9) below shows, so the many capital goods move together as a single factor with diminishing returns would, and Burmeister (1979, pp. 650–652) shows that an aggregate steady-state production function $F(\kappa)$ of an index of capital κ, with $F(\kappa)$ equal to net output per worker, $F'(\kappa) = r$ and F concave, exists if and only if the economy is regular. Because the theory does not restrict the available techniques, we have to resort to empirical evidence to see whether technology follows regularity, which we discuss in Section 5.2.

Regularity is only a necessary condition for the aggregate marginal productivity theory and is not enough on its own. The theory also identifies the rate of interest with the long-run marginal product of capital, and this identification requires the further condition $w'(r) = -k$ derived below. The long-run marginal product of capital compares long-run positions, so it is a total derivative across those positions and not a partial derivative. Take a stationary economy and measure all values in units of its net output. At every long-run position, the value of net output divides into wages and interest on the value of capital, $y = w + rk$, where y is net output per worker, w is the wage and k is the value of capital per worker. Across long-run positions, y, w and k are functions of the rate of

interest. Differentiating $y = w + rk$ with respect to r gives the long-run marginal product of capital, wherever $k'(r) \neq 0$:

$$\frac{dy}{dk} = \frac{y'(r)}{k'(r)} = r + \frac{w'(r) + k}{k'(r)}. \tag{7}$$

The long-run marginal product equals the rate of interest only when $w'(r) = -k$, that is, when the slope of the wage-interest frontier equals capital per worker. Samuelson (1962) constructed his surrogate production function from techniques with straight wage-interest frontiers, and in that case the condition holds. The condition fails in general because the prices that measure k change with the rate of interest, as Section 4 shows.

A second identity shows what the condition requires. Let $\mathbf{x}$ be the gross outputs per worker, so that $l \cdot \mathbf{x} = 1$, let $\mathbf{d}$ be the composition of net output, so that $\mathbf{x} - A^\top \mathbf{x} = y\,\mathbf{d}$, and let $\mathbf{K} = A^\top \mathbf{x}$ be the capital goods per worker. Let $\mathbf{P} = \mathbf{p}/(\mathbf{p} \cdot \mathbf{d})$ be the prices of capital goods in units of net output, so that $k = \mathbf{P} \cdot \mathbf{K}$. The change in capital per worker then splits into a real and a price Wicksell effect,

$$k'(r) = \mathbf{P} \cdot \frac{d\mathbf{K}}{dr} + \frac{d\mathbf{P}}{dr} \cdot \mathbf{K}, \tag{8}$$

where the first term is the change in the quantities of capital goods valued at current prices and the second is the change in the value of the existing capital goods. When techniques form a continuum, net output per worker satisfies

$$y'(r) = r\,\mathbf{P} \cdot \frac{d\mathbf{K}}{dr}. \tag{9}$$

To derive Equation (9), consider the change in techniques and quantities as the rate of interest changes, with the wage as numeraire. Each technique in use minimizes cost, so the change in technique leaves unit costs unchanged to first order, $(1 + r)\,dA\,\mathbf{p} + dl = 0$, and weighting by gross outputs gives $(1 + r)\,\mathbf{p} \cdot dA^\top \mathbf{x} = -\,dl \cdot \mathbf{x}$. Differentiating $l \cdot \mathbf{x} = 1$ gives $l \cdot d\mathbf{x} = -\,dl \cdot \mathbf{x}$. Substituting the price equation $\mathbf{p} = (1 + r)A\mathbf{p} + l$ into $\mathbf{p} \cdot d\mathbf{x}$ and using these two relations gives

$$\mathbf{p} \cdot d\mathbf{x} = (1 + r)\,\mathbf{p} \cdot A^\top d\mathbf{x} + l \cdot d\mathbf{x} = (1 + r)\,\mathbf{p} \cdot \left(A^\top d\mathbf{x} + dA^\top \mathbf{x}\right) = (1 + r)\,\mathbf{p} \cdot d\mathbf{K}.$$

Since $y\,\mathbf{d} = \mathbf{x} - \mathbf{K}$ with $\mathbf{d}$ fixed, $(\mathbf{p} \cdot \mathbf{d})\,dy = \mathbf{p} \cdot d\mathbf{x} - \mathbf{p} \cdot d\mathbf{K} = r\,\mathbf{p} \cdot d\mathbf{K}$, and dividing by $\mathbf{p} \cdot \mathbf{d}$ gives Equation (9). Burmeister (1979, p. 646) states the same relation with a single consumption good as numeraire, and Fratini (2013, p. 353) derives the same relation from the same cost-minimization argument and concludes that, at a positive rate of interest, the real Wicksell effect is negative if and only if net output per worker falls with the rate of interest.

Differentiating $y = w + rk$ gives $y'(r) = w'(r) + k + r\,k'(r)$, and substituting Equa-

tions (8) and (9) gives

$$w'(r) + k = -\,r\,\frac{d\mathbf{P}}{dr} \cdot \mathbf{K}. \tag{10}$$

At a positive rate of interest, the condition $w'(r) = -k$ therefore holds if and only if the price Wicksell effect is zero, that is, if a change in the rate of interest leaves the value of the existing capital goods unchanged. The only realistic case in which the value stays unchanged is one where the prices of capital goods do not change with the rate of interest. Burmeister's index κ does not remove this condition. The index is built so that its change equals the real Wicksell effect, which gives $F'(\kappa) = r$, but the index is a different quantity from the value of capital k. A marginal productivity account of distribution pays capital its marginal product times its quantity, which is $r\kappa$ for the index, while the income that capital actually receives is rk, so that $y = w + rk$. The two payments coincide over a range of rates of interest only if the index and the value of capital move together, that is, only if the price Wicksell effect is zero, which by Equation (10) is again the condition $w'(r) = -k$. Burmeister (1979, p. 652) also notes that the index involves unobservable economic variables. Nuti (1979, p. 653) makes an identical point: when capital is measured as a value, the rate of profit embodied in the prices used to aggregate capital goods may differ from the derivative of output with respect to the value of capital.

5.2 Evidence from input-output data

Quantitative DSGE models measure capital as a single aggregate and derive output from an aggregate production function. In such a model the wage and the rate of interest are the marginal products of labor and capital, and the condition $w'(r) = -k$ holds by construction. Evidence on long-run wage-interest frontiers therefore tests whether DSGE aggregation is valid, even though the models are not long-run theories. Han and Schefold (2006) compute such frontiers from the input-output tables of nine OECD countries, including the United States. Most switch points on their frontiers are regular, so the aggregate relation between the rate of interest and capital per worker holds approximately. Along every individual wage-interest curve, however, the value of capital changes with the rate of interest, so the condition $w'(r) = -k$ fails to some degree in every case. Flaschel et al. (2012) object that mixing processes across countries and times is illegitimate once capital stocks are taken into account. Schefold (2012) defends input-output tables as representing states of technical knowledge.

The failure of $w'(r) = -k$ should be expected. By Equation (10), the condition $w'(r) = -k$ holds if and only if a change in the rate of interest leaves the value of the existing capital goods unchanged, and the only realistic case in which this happens is one where the prices of capital goods do not change, so almost any data should reject the condition. Aggregation can hold approximately while the long-run marginal product of

capital still differs from the rate of interest. Normative conclusions drawn from aggregate models, such as the claim that distribution by marginal product is just, cannot be trusted.

6 Conclusion

The Cambridge capital controversy is usually told as a contest between two positions. The UK side held that neoclassical theory cannot support long-run analysis of capital and distribution. The US side, and the general equilibrium theorists who followed, held that the move to Walrasian and then Radner equilibrium settled the matter. The neoclassical theory supplies a short-run center of gravitation, given in practice by the policy rate of the central bank, so a dispute about whether a center of gravitation exists cannot by itself settle the controversy. Garegnani's observation that intertemporal equilibria lack persistence (Garegnani, 1976; Petri, 2004) concerns the long-run center of gravitation and leaves the short-run center of gravitation given by the money rate unaffected. This paper proposes reading the controversy as a question of whether that short-run center of gravitation, together with the short-run justification of neoclassical capital theory, is enough, or whether the short-run theory needs to aggregate into a long-run theory. Neoclassical capital theory has a short-run part, made of equilibrium conditions at given endowments and prices, and a long-run part, made largely of comparative statics. The short-run part describes individual choices, while the long-run part describes how a society views its economy as a whole. The short-run part is built on partial derivatives, justified by price-taking. At given prices each firm's demand for capital slopes downward, and if this short-run justification is accepted as sufficient, the general-equilibrium reply establishes consistency of the neoclassical approach in the sense Hahn (1982a) defended, with the long-run aggregate theory left to empirical derivation. The long-run theory then survives as an approximation in positive economics where technology is regular, while its use in normative economics requires the restrictive condition $w'(r) = -k$, which the evidence rejects.

The Cambridge capital controversy therefore shows that, for the question of long-run distribution, a free market is not enough even for those who accept that distribution by marginal product is just. This point should not be controversial. Distinguishing the short-run center of gravitation from the long-run center of gravitation should help avoid the cross-talk that has run through decades of the debate.

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References

Anderson, C. M., Plott, C. R., Shimomura, K.-I., and Granat, S. (2004). Global instability in experimental general equilibrium: The Scarf example. *Journal of Economic Theory*, 115(2):209–249.

Arrow, K. J. and Debreu, G. (1954). Existence of an equilibrium for a competitive economy. *Econometrica*, 22(3):265–290.

Athreya, K. B. (2013). *Big Ideas in Macroeconomics: A Nontechnical View*. MIT Press, Cambridge, MA.

Aumann, R. J. (1964). Markets with a continuum of traders. *Econometrica*, 32(1/2):39–50.

Burmeister, E. (1979). Professor Pasinetti's 'unobtrusive postulate', regular economies, and the existence of a well-behaved aggregate production function. *Revue d'Economie Politique*, 89(5):644–652.

Burmeister, E. (1980). *Capital Theory and Dynamics*. Cambridge University Press, Cambridge.

Clark, J. B. (1899). *The Distribution of Wealth: A Theory of Wages, Interest and Profits*. Macmillan, New York.

Debreu, G. (1959). *Theory of Value: An Axiomatic Analysis of Economic Equilibrium*. Yale University Press, New Haven.

Debreu, G. (1974). Excess demand functions. *Journal of Mathematical Economics*, 1(1):15–21.

Debreu, G. and Scarf, H. (1963). A limit theorem on the core of an economy. *International Economic Review*, 4(3):235–246.

Diamond, P. A. (1965). National debt in a neoclassical growth model. *American Economic Review*, 55(5):1126–1150.

Fisher, F. M. (1983). *Disequilibrium Foundations of Equilibrium Economics*. Cambridge University Press, Cambridge.

Flaschel, P., Franke, R., and Veneziani, R. (2012). The measurement of prices of production: An alternative approach. *Review of Political Economy*, 24(3):417–435.

Fratini, S. M. (2013). Real Wicksell effect, demand for capital and stability. *Metroeconomica*, 64(2):346–360.

Gale, D. (1986). Bargaining and competition part I: Characterization. *Econometrica*, 54(4):785–806.

Garegnani, P. (1970). Heterogeneous capital, the production function and the theory of distribution. *The Review of Economic Studies*, 37(3):407–436.

Garegnani, P. (1976). On a change in the notion of equilibrium in recent work on value and distribution: A comment on Samuelson. In Brown, M., Sato, K., and Zarembka, P., editors, *Essays in Modern Capital Theory*, pages 25–45. North-Holland, Amsterdam.

Garegnani, P. (2000). Savings, investment and capital in a system of general intertemporal equilibrium. In Kurz, H. D., editor, *Critical Essays on Piero Sraffa's Legacy in Economics*, pages 392–445. Cambridge University Press, Cambridge.

Garegnani, P. (2005). Capital and intertemporal equilibria: A reply to Mandler. *Metroeconomica*, 56(4):411–437.

Giraud, G. (2003). Strategic market games: An introduction. *Journal of Mathematical Economics*, 39(5–6):355–375.

Goeree, J. K. and Lindsay, L. (2016). Market design and the stability of general equilibrium. *Journal of Economic Theory*, 165:37–68.

Hahn, F. (1982a). The neo-Ricardians. *Cambridge Journal of Economics*, 6(4):353–374.

Hahn, F. (1982b). Stability. *Handbook of Mathematical Economics*, 2:745–793.

Han, Z. and Schefold, B. (2006). An empirical investigation of paradoxes: Reswitching and reverse capital deepening in capital theory. *Cambridge Journal of Economics*, 30(5):737–765.

Hicks, J. R. (1939). *Value and Capital: An Inquiry into Some Fundamental Principles of Economic Theory*. Clarendon Press, Oxford.

Jorgenson, D. W. (1963). Capital theory and investment behavior. *The American Economic Review*, 53(2):247–259.

Magill, M. and Shafer, W. (1991). Incomplete markets. In Hildenbrand, W. and Sonnenschein, H., editors, *Handbook of Mathematical Economics*, volume 4, pages 1523–1614. North-Holland, Amsterdam.

Makowski, L. and Ostroy, J. M. (2001). Perfect competition and the creativity of the market. *Journal of Economic Literature*, 39(2):479–535.

Mandler, M. (2005). Well-behaved production economies. *Metroeconomica*, 56(4):477–494.

Mas-Colell, A., Whinston, M. D., and Green, J. R. (1995). *Microeconomic Theory*. Oxford University Press, New York.

Nuti, D. M. (1979). The irrelevance of reswitching: A reply to Pasinetti. *Revue d'Economie Politique*, 89(5):652–654.

Opocher, A. and Steedman, I. (2015). *Full Industry Equilibrium: A Theory of the Industrial Long Run*. Cambridge University Press, Cambridge.

Pasinetti, L. L. (1979). The 'unobtrusive postulate' of neoclassical economic theory: A rejoinder to professor Burmeister. *Revue d'Economie Politique*, 89(5):654–656.

Petri, F. (2004). *General Equilibrium, Capital and Macroeconomics: A Key to Recent Controversies in Equilibrium Theory*. Edward Elgar, Cheltenham.

Radner, R. (1972). Existence of equilibrium of plans, prices, and price expectations in a sequence of markets. *Econometrica*, 40(2):289–303.

Robinson, J. (1953). The production function and the theory of capital. *The Review of Economic Studies*, 21(2):81–106.

Saari, D. G. and Simon, C. P. (1978). Effective price mechanisms. *Econometrica*, 46(5):1097–1125.

Samuelson, P. A. (1962). Parable and realism in capital theory: The surrogate production function. *The Review of Economic Studies*, 29(3):193–206.

Samuelson, P. A. (1966). A summing up. *The Quarterly Journal of Economics*, 80(4):568–583.

Scarf, H. (1960). Some examples of global instability of the competitive equilibrium. *International Economic Review*, 1(3):157–172.

Schefold, B. (2005). Reswitching as a cause of instability of intertemporal equilibrium. *Metroeconomica*, 56(4):438–476.

Schefold, B. (2012). A comment on 'the measurement of prices of production' by peter flaschel, reiner franke and roberto veneziani. *Review of Political Economy*, 24(3):437–444.

Shapley, L. S. and Shubik, M. (1977). Trade using one commodity as a means of payment. *Journal of Political Economy*, 85(5):937–968.

Shephard, R. W. (1953). *Cost and Production Functions*. Princeton University Press, Princeton, NJ.

Sonnenschein, H. (1973). Do Walras' identity and continuity characterize the class of community excess demand functions? *Journal of Economic Theory*, 6(4):345–354.

Sraffa, P. (1932). Dr. Hayek on Money and Capital. *The Economic Journal*, 42(165):42–53.

Sraffa, P. (1960). *Production of Commodities by Means of Commodities: Prelude to a Critique of Economic Theory*. Cambridge University Press, Cambridge.

Steedman, I. (2005). Long run input use–input price relations and the cost function hessian. Working paper, Manchester Metropolitan University.

Varian, H. R. (1984). The nonparametric approach to production analysis. *Econometrica*, 52(3):579–598.